The Cyber Control Tower (CCT)

Track:

Infrastructure and Security - Cyber Security Incident Response Analyst

Instructor:

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Team Members:

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- **Ali Ahmed Ali**

1. Project Planning & Management

1.1 Project Proposal

Overview

The Cyber Control Tower (CCT) is a web-based cyber security platform that simulates a real-world Security Operations Center (SOC).
It integrates log ingestion, incident correlation, MITRE ATT&CK mapping, human risk scoring, and performance metrics (MTTD & MTTR) into a unified dashboard.

Objectives

- Transform raw logs into meaningful security insights
- Detect and correlate cyber attacks in real time
- Map incidents to the MITRE ATT&CK framework
- Evaluate human risk using Trust Score
- Measure SOC performance using MTTD and MTTR
- Visualize the organization as a Cyber Digital Twin

Scope

Includes:

- Log simulation (Windows / Linux / Authentication logs)
- Graph-based digital twin visualization
- Incident detection and correlation
- MITRE mapping
- Trust Score engine
- Dashboard visualization

Excludes:

- Real-world deployment
- Full SIEM integration

1.2 Project Plan

Timeline

Week	Task
1	Planning and Requirements

Week	Task
2	Data Simulation and Backend Setup
3	Cyber Digital Twin (Graph Layer)
4	Correlation Engine and MITRE Mapping
5	Trust Score and Metrics
6	Frontend Dashboard
7	Integration
8	Testing and Presentation

Milestones

- Data ingestion completed
- Graph visualization implemented
- Incident detection operational
- Dashboard finalized

Deliverables

- Source code
- Final report
- Presentation
- Demonstration scenario

1.3 Task Assignment & Roles

Role	Responsibility
Team Lead	System design and coordination
Backend Developer 1	Log ingestion and APIs
Backend Developer 2	Correlation and MITRE mapping
Data/Logic Developer	Trust Score and MTTD/MTTR
Frontend Developer 1	Dashboard interface
Frontend Developer 2	Graph visualization

1.4 Risk Assessment & Mitigation Plan

Risk	Mitigation
System complexity	Modular design
Limited time	Prioritize core features
Integration challenges	Early integration testing
Data inconsistency	Use structured JSON format

1.5 KPIs (Key Performance Indicators)

- Mean Time to Detect (MTTD)
- Mean Time to Respond (MTTR)
- Number of detected incidents
- Correlation accuracy
- System uptime
- Dashboard usage and interaction

2. Literature Review

Topics Covered

- Security Operations Centers (SOC)
- SIEM systems such as Splunk and Wazuh
- MITRE ATT&CK framework
- Graph-based security modeling
- Behavioral analysis and insider threat detection

Key Insight

Traditional security systems focus primarily on log collection and monitoring, while the proposed system emphasizes behavioral analysis, attack correlation, and human risk evaluation combined with visual representation.

Note:

Feedback, suggested improvements, and grading criteria are typically added after project evaluation.

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3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Needs
Security Analyst	Monitor and analyze threats
SOC Manager	Evaluate system performance
IT Administrator	Understand system impact
Instructor	Assess project quality

3.2 User Stories

- As a Security Analyst, I want to monitor incidents in real time so that I can respond quickly
- As a SOC Manager, I want to track MTTD and MTTR to evaluate performance
- As a user, I want my Trust Score to reflect my behavior

3.3 Use Cases

- View Cyber Digital Twin
- Detect and analyze incidents
- Identify MITRE stages
- Monitor Trust Scores
- Simulate attack scenarios

3.4 Functional Requirements

- Log ingestion system
- Incident detection engine
- MITRE mapping module
- Trust Score calculation
- Interactive dashboard
- Attack simulation module

3.5 Non-Functional Requirements

- Performance: real-time or near real-time updates
- Security: data isolation and integrity

- Usability: clear and intuitive interface
- Reliability: stable and consistent system behavior

1. System Analysis & Design

1.1 Problem Statement & Objectives

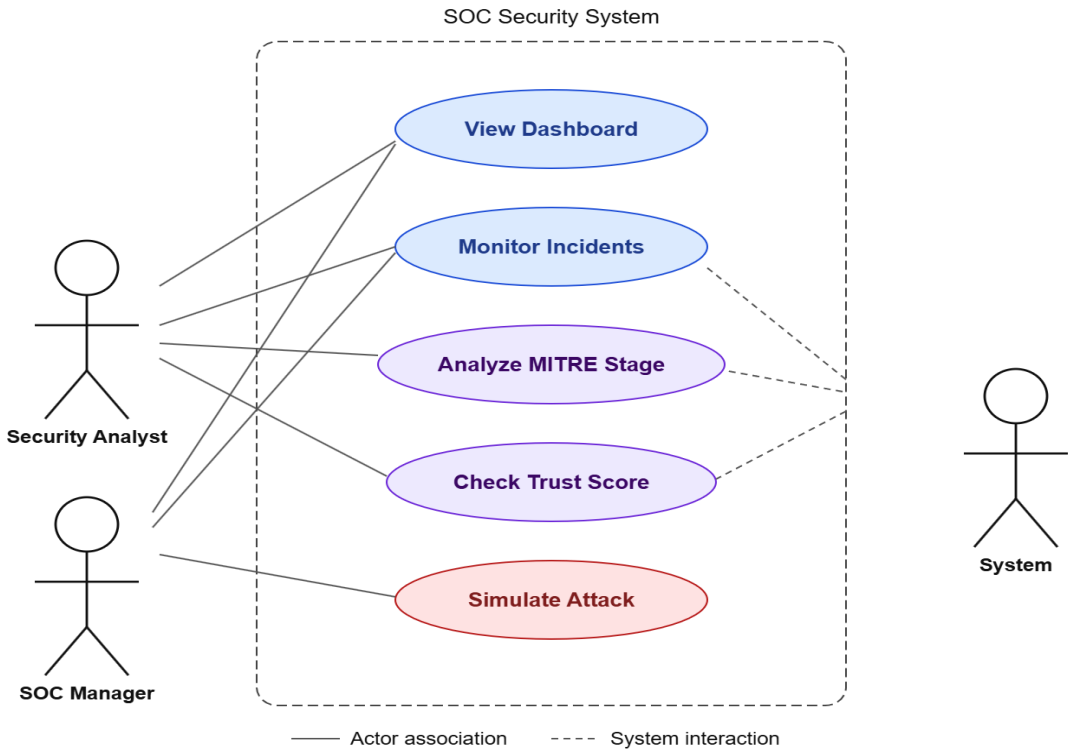
Problem statement :

Traditional security systems generate large volumes of logs without providing clear insights or relationships between events.

This makes it difficult for security analysts to detect attacks quickly, understand their progression, and respond effectively.

Objectives:

- Correlate logs into meaningful security incidents
- Visualize the system as a Cyber Digital Twin
- Map attacks to MITRE ATT&CK
- Evaluate user behavior using Trust Score
- Measure performance using MTTD and MTT1.2 Use Case Diagram & Descriptions



1.3 Functional & Non-Functional Requirements

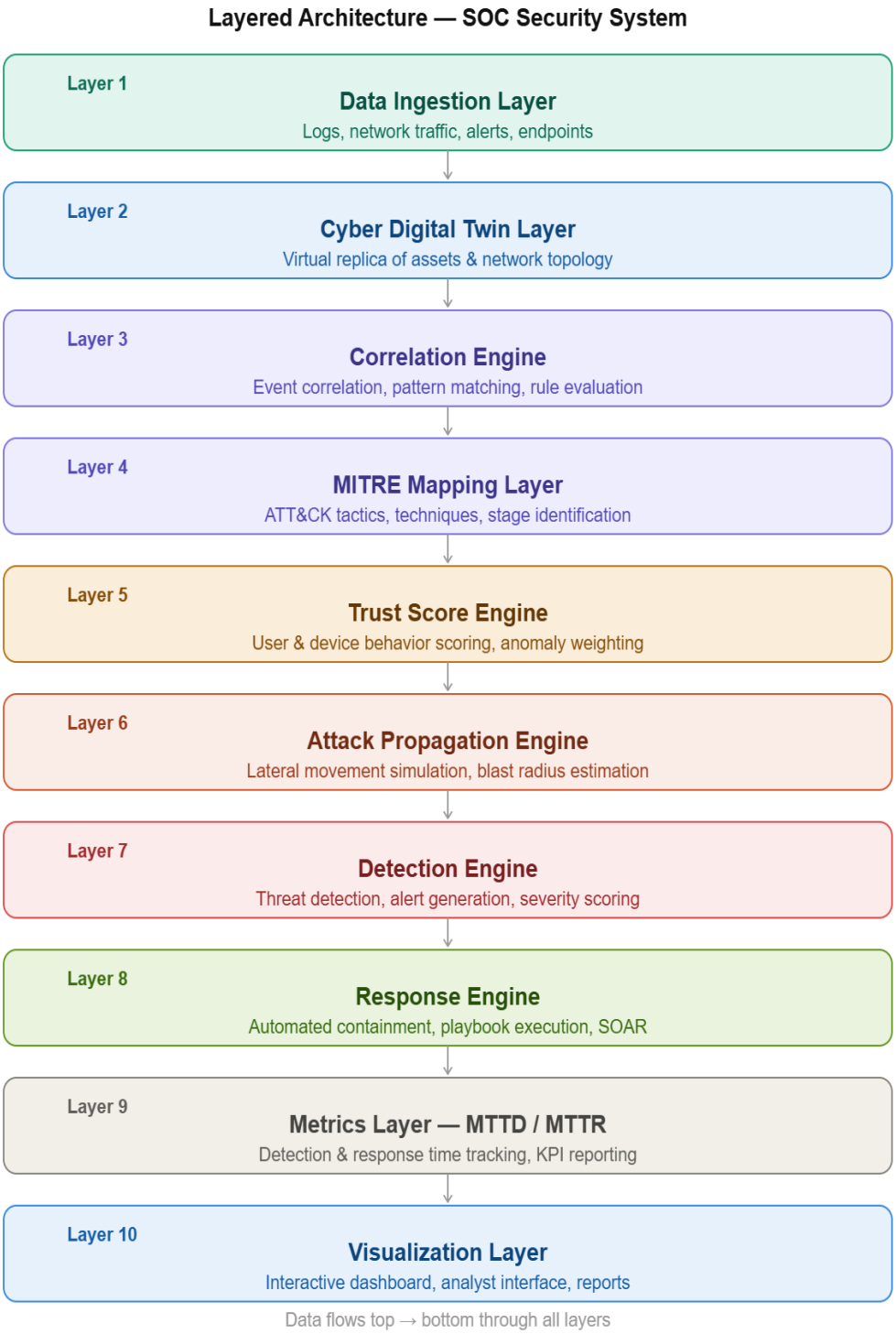
Functional:

- Log ingestion
- Event correlation
- MITRE mapping
- Trust Score calculation
- Incident detection
- Dashboard visualization

Non-Functional:

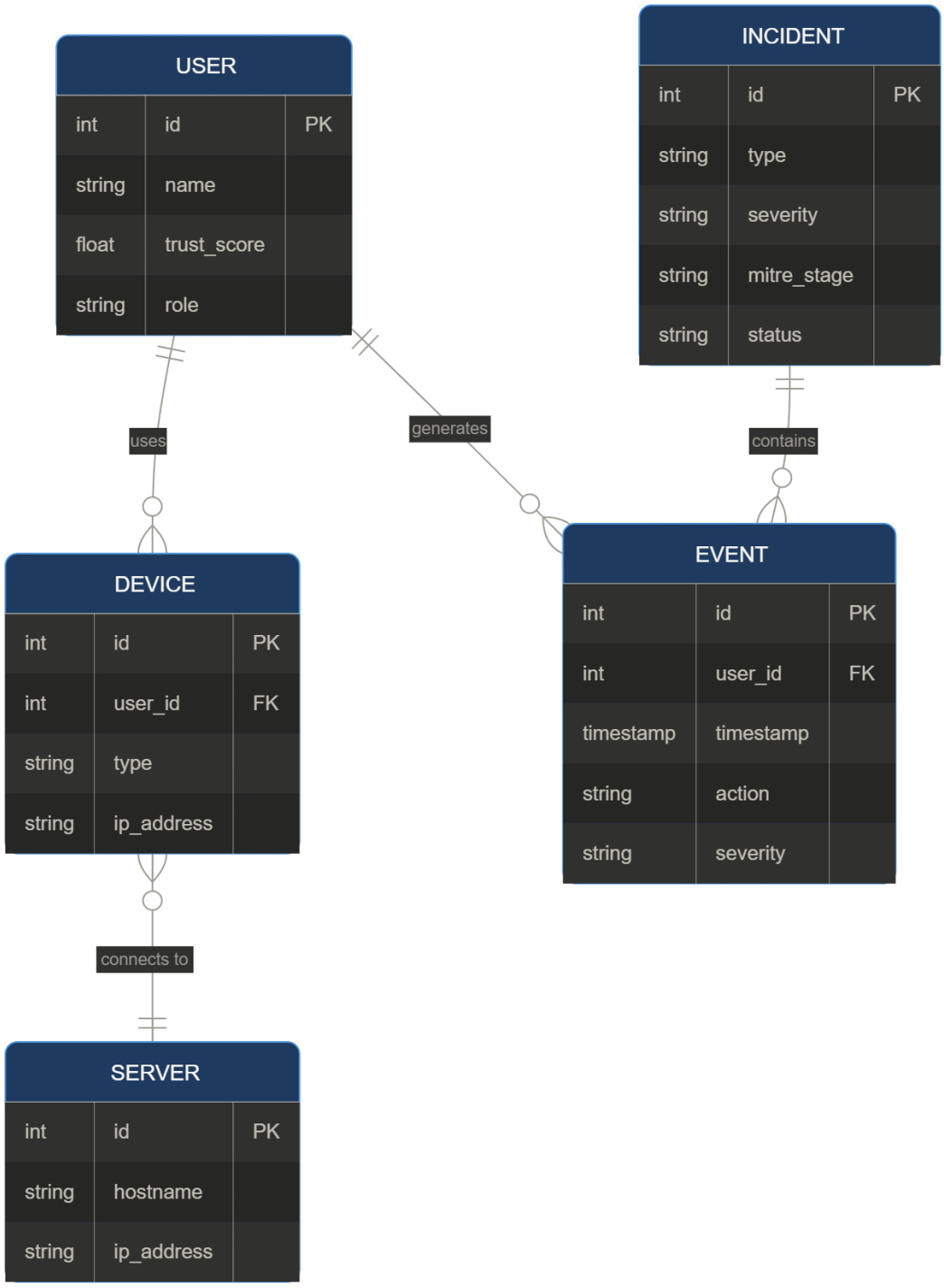
- Real-time performance
- System reliability
- Data integrity
- Usability

1.4 Software Architecture



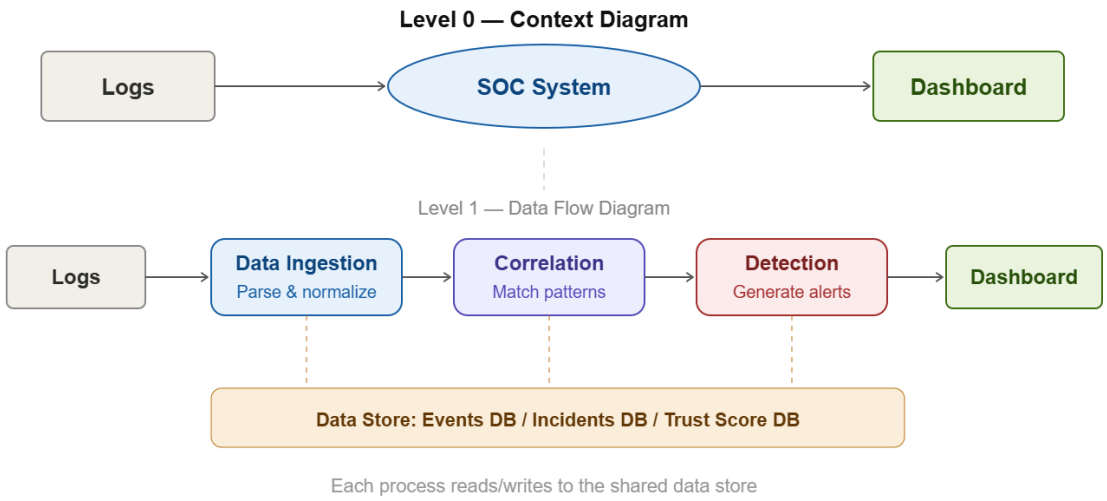
2. Database Design & Data Modeling

2.1 ER Diagram

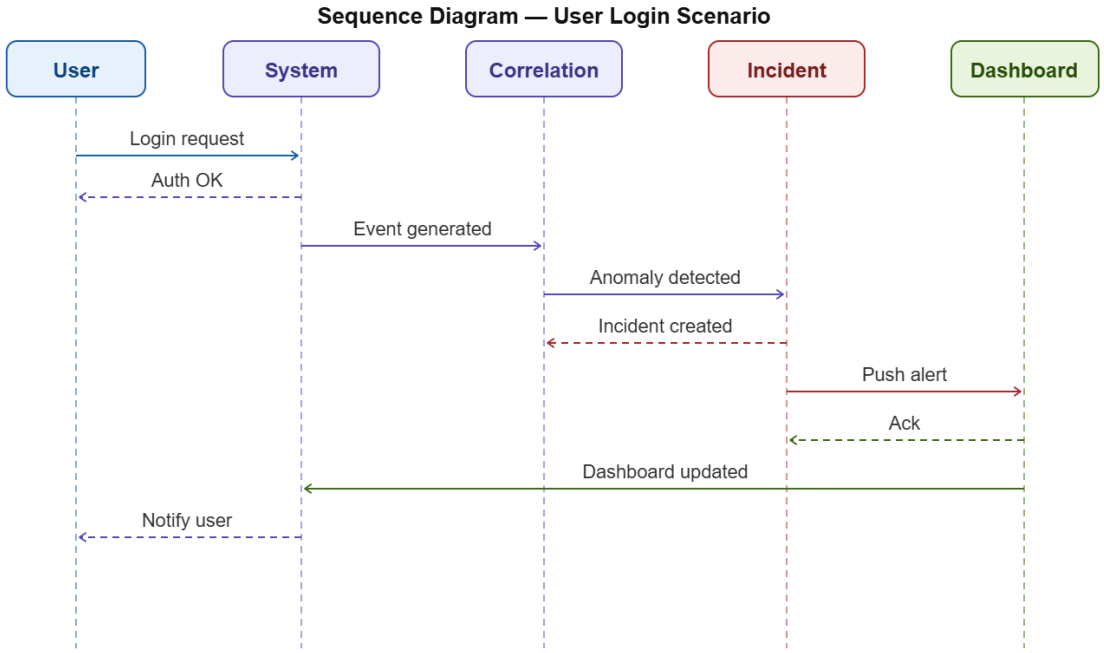


3. Data Flow & System Behavior

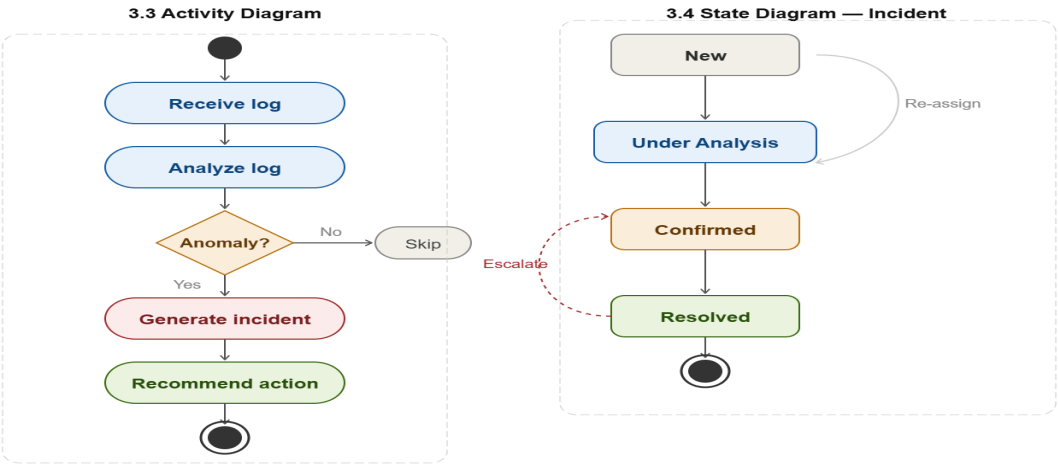
3.1 DFD



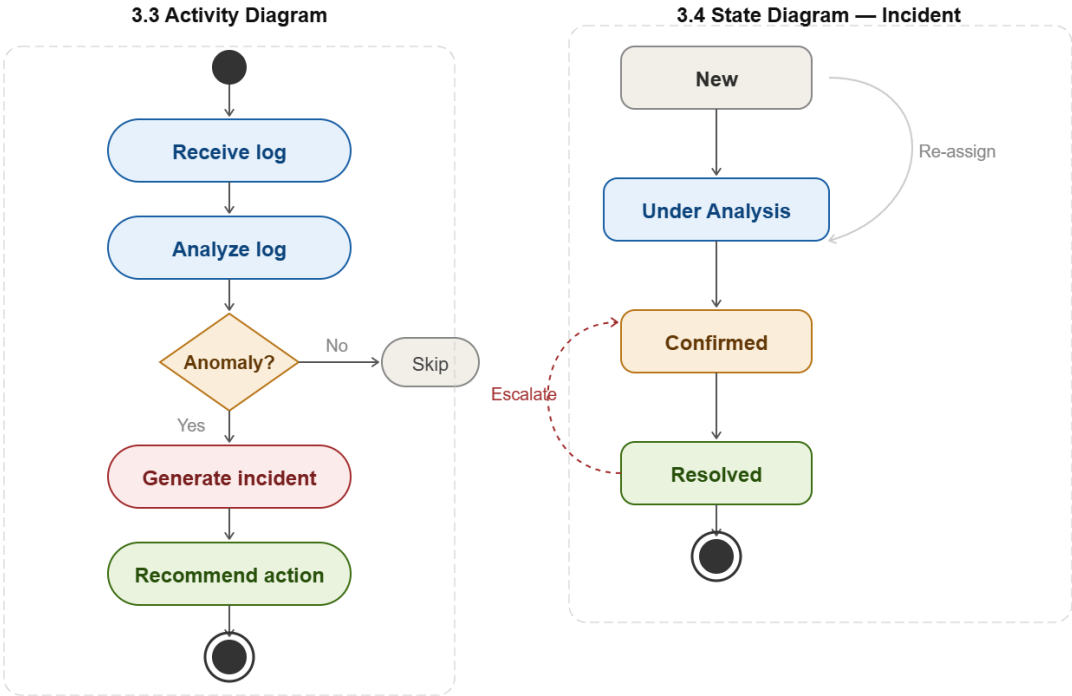
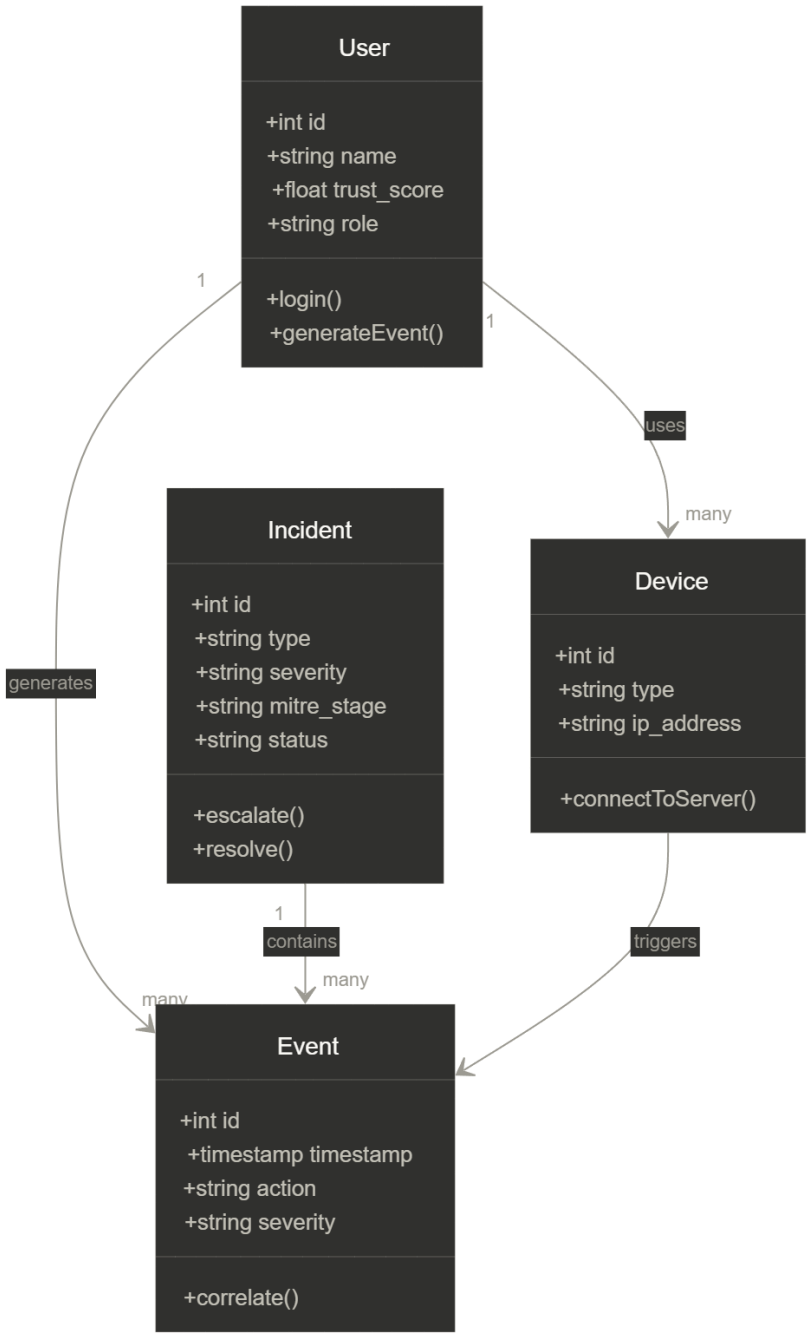
3.2 Sequence Diagram



3.3 Activity Diagram



3.3 Activity Diagram & 3.4 State Diagram



3.5 Class Diagram